Chapter 20

Outcomes Research

Erin R. Weeda[1], Nicole A. Pilch[1] and Lieven Annemans[2]
[1]Medical University of South Carolina, Charleston, SC, United States; [2]Ghent University, Ghent, Belgium

Learning Objectives:

Objective 20.1 Define and classify outcomes research.
Objective 20.2 Contrast outcomes research with traditional clinical trial research.
Objective 20.3 Define and explain how to measure health-related quality of life and patient-reported outcomes.
Objective 20.4 Generalize how outcomes research is gaining importance in clinical practice.

OBJECTIVE 20.1. DEFINE AND CLASSIFY OUTCOMES RESEARCH

Patients, clinicians, payers, policy-makers, and the society at large look forward to evidence that novel medications and technologies are beneficial and efficiently implemented in real-world settings. Outcomes research (OR), which is a growing field that combines principles of epidemiology, clinical research, health economics, quality of life assessment, and health policy, is critical in obtaining this evidence.[1] Whereas clinical trials aim to generate knowledge on safety and efficacy, OR studies the benefits, risks, and costs that guide stakeholders in a real-world environment. OR has a focus on patient perspectives and considers the value (e.g., benefits achieved) for money and efforts spent. Patient-reported outcomes (PROs) and clinician-reported outcomes are components of OR.

Because the term OR applies to many research activities, it lacks a consistent definition. This chapter uses the definition proposed by the United States (US) Agency for Healthcare Research and Quality (AHRQ), which is one of the main federal agencies dedicated to improving the safety and quality of healthcare delivery in the US.[2] AHRQ defines OR as follows:

> *Outcomes research seeks to understand the end results of particular healthcare practices and interventions. End results include effects that people experience and care about, such as change in the ability to function. In particular, for individuals with chronic conditions—where cure is not always possible—end results include quality of life as well as mortality. By linking the care people get to the outcomes they experience, outcomes research has become the key to developing better ways to monitor and improve the quality of care.*

According to the AHRQ definition, OR focuses on the "end result" of a healthcare intervention.[2] This end result is measured after the intervention has been implemented in clinical practice, meaning the treatment benefit is assessed in the presence of real-world factors (e.g., patient adherence, access to the intervention, clinician prescribing patterns, etc.). In other words, OR measures effectiveness, or how well the treatment works in clinical practice, rather than the efficacy, or how well the treatment works in an ideal, controlled environment like a traditional clinical trial.[3]

OR has also been defined as *any research that attempts to measure the effectiveness, efficiency, equality, and patient-centeredness of healthcare*.[4]

Characteristics of Outcomes Research

- Focuses on the end result of a healthcare intervention.
- Measures final health outcomes rather than surrogate outcomes.
- Common research method in clinical practice.

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- High applicability (granted confounders are considered when interpreting the results) with limitations that are similar to other observational studies.
- Measures quality of care in comparable healthcare systems. Healthcare institutions may use OR to evaluate the quality of care that they are providing in comparison to the care provided by other institutions.
- Considers viewpoints of multiple stakeholders, including patients, policy-makers, providers, payers, and the pharmaceutical industry.
- To capture the perspectives of stakeholders such as payers and the pharmaceutical industry, OR may involve economic analyses. The interdependent relationship between OR and economics led to the development of the International Society of Pharmacoeconomics and Outcomes Research (ISPOR) in 1995. Information and guidance disseminated by ISPOR are highly regarded resources for OR.[5]
- Shares methodology with pharmacoepidemiology (which studies the use of medications at the population level).[6]
- As opposed to the controlled environment of a clinical trial, OR measures the impact of a medication when real-world factors are introduced.

Economic, Clinical, and Humanistic Outcomes

The three main outcomes measured by OR are as follows: clinical, economic, and humanistic (Fig. 20.1).[7,8] Clinical outcomes are medical events (e.g., the rate of myocardial infarction or hospitalization due to heart failure) or physiologic measures (e.g., cholesterol, blood pressure), whereas economic outcomes are disease- or treatment-related resource use and costs. Humanistic outcomes capture the impact of a treatment on patient well-being or health-related quality of life (HRQoL). The interplay between these outcomes is described by the economic, clinical, and humanistic outcomes (ECHO) model.[7,8] Medical research has traditionally relied on the measurement of clinical outcomes; therefore, the ECHO model is unique in that it allows for a more comprehensive approach to assessing the impact of an intervention.

Donabedian's Structure–Process–Outcome Model

Donabedian's structure–process–outcome (SPO) model is used in outcomes studies that are designed to measure or determine the impact of an intervention on quality of care, most commonly at a single institution or within the health system.[9] The model consists of three domains: structure, process, and outcome (Fig. 20.2).

Structure represents the characteristics of the healthcare environment. It consists of the resources that are used to provide care (i.e., the equipment, facilities, and staff). An example of a structural issue would be if a patient was unable to undergo diagnostic imaging because the institution does not have the needed imaging equipment.

Process describes the proportion of patients receiving optimal treatment. For example, it is considered standard practice for patients contacting emergency medical services (EMS) with chest pain to receive an electrocardiogram (ECG) prior to hospital arrival.[10] If only 10% of patients with chest pain transported to the hospital by EMS received an ECG, this could be a process issue.

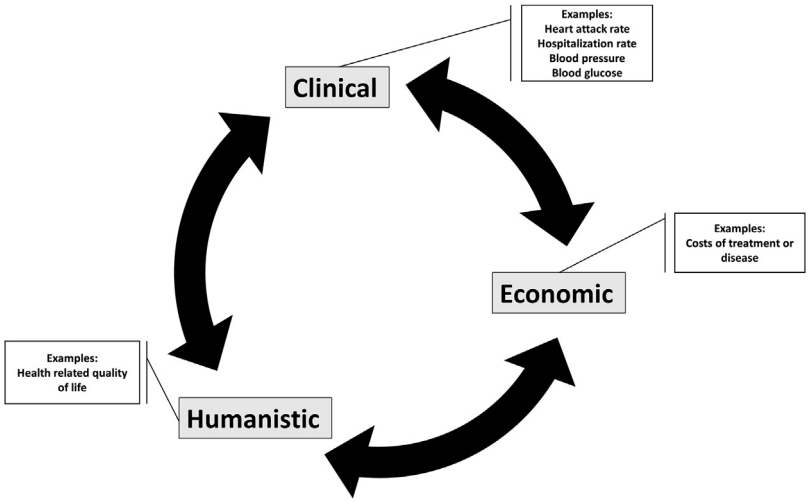


FIGURE 20.1 The economic, clinical, and humanistic outcomes model.

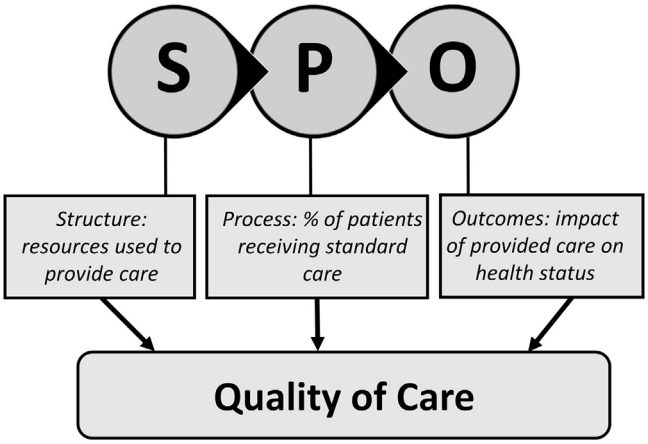


FIGURE 20.2 Donabedian's structure–process–outcome (SPO) model.

Outcomes represent the impact of care on patient health status. These can be measured using the outcomes described in the ECHO model. OR attempts to evaluate and find ways to optimize all three domains in the SPO model to increase quality of care.[9]

Examples of Outcomes Research

Using the above definitions, studies that can be classified as OR span a wide spectrum of clinical areas.[2,4,7,9] An outcomes study might evaluate the rate of hospital-acquired infections after an intervention to increase handwashing among employees at a single hospital. Studies comparing length of stay and costs associated with the use of two antibiotics using billing data from multiple hospitals can also be classified as OR. Other examples include measuring patients' HRQoL while they undergo treatment for cancer, estimating the proportion of patients that achieve their blood pressure goals after receiving education from a pharmacist or calculating 30-day readmission rates among patients that received a novel surgical procedure.

OBJECTIVE 20.2. CONTRAST OUTCOMES RESEARCH WITH TRADITIONAL CLINICAL TRIAL RESEARCH

It has been suggested that "if we want more evidence-based practice, we need more practice-based evidence.[11]" Clinical trials are essential to provide regulatory approval for new medicines. However, these trials use strict selection criteria (e.g., excluding patients with extensive comorbidities) and careful monitoring; thus it is never guaranteed that the observed benefit will hold true in the less-controlled, real-world environment of clinical practice. OR provides a means of measuring the impact of a medication (or other treatment) when real-world factors are introduced.

Overlap Between Outcomes Reasearch and Clinical Trials

As the field of OR progresses, there is increasing overlap between OR and clinical trials. Clinical trials are now incorporating measures that were traditionally components of OR into study protocols. For instance, many clinical trials measure HRQoL as a secondary endpoint.[12] Furthermore, the use of pragmatic trials is increasing, which maintain some qualities of clinical trials (e.g., randomly assigning patients to treatment groups). However, these trials also incorporate methodology used in OR to increase applicability, including the use of less stringent inclusion criteria and simplified study design.[13] An example is "A Pragmatic Trial to Evaluate the Comparative Effectiveness Between Dapagliflozin and Standard of Care in Type 2 Diabetes Patients (DECIDE Study)."[14] Much like a traditional clinical trial, the study design uses randomization to assign patients to either dapagliflozin 10 mg (a glucose-lowering medication for diabetes) or a standard of care arm. Unlike a traditional clinical trial, clinicians are free to replace dapagliflozin with other glucose-lowering drug(s) or use additional medications without restrictions. Patients and the clinicians caring for them are also aware of treatment status (i.e., who receives dapagliflozin and who receives the standard of care). In contrast, patients and clinicians are often unaware of treatment status (i.e., they are "blinded") in traditional clinical trials, as this reduces bias.

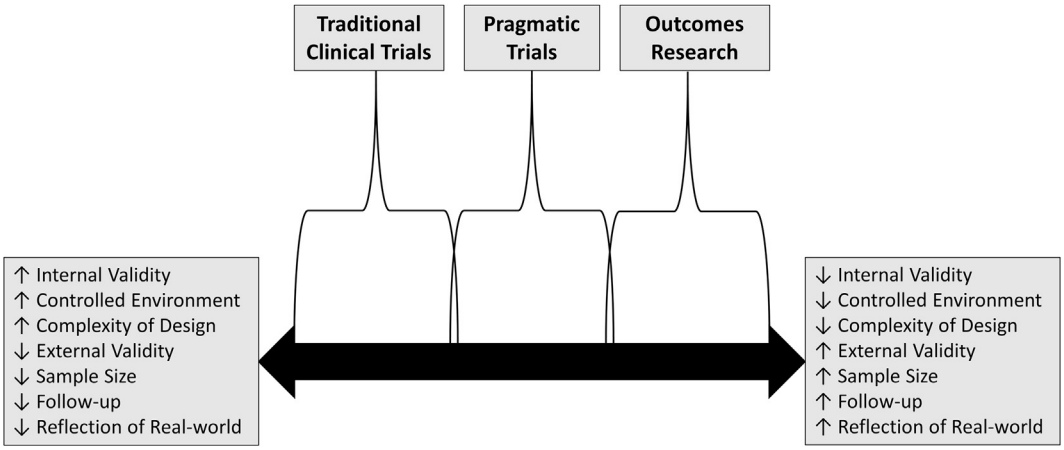


FIGURE 20.3 Relationship between traditional clinical trials, pragmatic trials, and outcomes research.

The relationship between traditional clinical trials, pragmatic trials, and OR is shown in Fig. 20.3.[13] However, for the remaining of this section, we will contrast OR with the objectives and methods traditionally used in clinical trial research.

OR is best distinguished from traditional clinical trials by the questions it attempts to answer (Table 20.1).[3] Traditional clinical trials are designed to show the efficacy of an intervention, often for regulatory purposes.[3] In contrast, OR aims to show that the intervention truly improves health or quality of care in clinical practice. In clinical trial research, endpoints are determined by the viewpoints of clinicians and investigators who try to measure prognosis. In OR, outcomes are determined by taking into account the viewpoints of multiple stakeholders (e.g., patients, providers, and payers). To increase the validity of the study design (i.e., internal validity), clinical trials randomly assign patients to an intervention or control (e.g., placebo) group. In OR, the researchers observe outcomes as patients are treated according to standard practice (i.e., clinicians in discussion with the patients decide on treatment).

TABLE 20.1 Characteristics of Traditional Clinical Trials Contrasted With Outcomes Research

Traditional Clinical Trials	Outcomes Research
Efficacy: Aims to show that the intervention is effective, often to obtain regulatory approval	**Effectiveness:** Aims to show that the intervention improves the quality of care that patients receive and can be delivered in an efficient manner, often within a health system
Outcomes determined by the viewpoints of **clinicians and investigators** who try to select the most valid measure of disease	Outcomes determined by the viewpoints of **multiple stakeholders,** including patients, policy-makers, payers, etc.
Interventional: Patients assigned to treatment groups	**Observational:** Patients treated according to usual clinical practice
Higher **internal validity**	Lower internal validity
Lower applicability	Higher **applicability**
Prospective data collection (i.e., data collection planned prior to the start of the study)	**Data** often **collected from information designed for treatment** rather than research purposes (e.g., billing data or medical charts)
Measure impact of treatment in a **controlled environment** with extensive monitoring	Measure impact of treatment with monitoring **used in routine practice**
Can **exclude patients with extensive comorbid conditions** or medications that may make it unsafe for the patient to receive the intervention	Can **include patients regardless of comorbidity status**
Can **exclude those unlikely to adhere** to therapy	Can **include patients regardless of adherence status**
Limited sample size	**Larger sample size**
Shorter follow-up	**Longer follow-up**

TABLE 20.2 Examples of Surrogate Versus Final Health Outcomes

Surrogate Outcome	Final Health Outcome
Blood glucose control	Microvascular complications of diabetes (e.g., decreased kidney function) prevented
Blood pressure control	Cerebrovascular disease (e.g., stroke) prevented
Controlled lipid profile	Coronary heart disease (e.g., myocardial infarction) prevented
Normal white blood cell count	Cured infection
Elevated CD4 count	Opportunistic infections among patients with human immunodeficiency virus prevented

In clinical trials, internal validity is also increased because investigators decide what data will be collected prior to the start of the study.[3] This allows investigators to plan patient follow-up visits, mostly with a strict follow-up scheme, as well as collect laboratory values as frequently as needed, even if these measures are not routinely collected in clinical practice. In contrast, data used in outcomes studies are often collected from information designed for treatment rather than research purposes. OR may rely on retrospective data already available within medical charts or billing records. Because of this, the researchers do not have access to all applicable outcomes or baseline characteristics.

Clinical trials are also conducted in a more controlled environment than OR.[3] The clinical trial study protocol may require more interaction between patients and physicians (for reasons such as monitoring adverse events) than what is typical in routine care. Furthermore, in an attempt to maximize safety, patients with extensive comorbidities can be excluded from studies of interventions that are not proven to be safe. Those unlikely to adhere to therapy may also be excluded. Because researchers are not intervening in OR, they may include a broader group of patients (e.g., those with extensive comorbidities or poor adherence) who more closely mirror those encountered in clinical practice. Outcomes studies may therefore produce results that can be readily achieved in a healthcare setting.

In contrast to OR, some clinical trials utilize indirect measures of final health outcomes (e.g., laboratory values) referred to as surrogate outcomes (Table 20.2).[15] Because clinical trials are costly to conduct, surrogate outcomes are appealing because they are less costly and easier to measure (e.g., they can be obtained more quickly) than final health outcomes and can often be collected within a relatively short time frame. However, changes in surrogate outcomes do not always correspond to changes in final health outcomes. As OR can measure final health outcomes in an efficient manner, it can be used to confirm that an intervention improves surrogate outcomes and also improves final health outcomes.

OR can play an important role in evaluating the safety of a drug.[3] In clinical trials, a small proportion of patients are exposed to medications for short periods due to cost restraints. This may result in an incomplete understanding of the safety profile of a medication. Perhaps the most notorious example of a medication causing harmful side effects is the thalidomide crisis in Europe, Japan, and Australia.[16] In the 1950s, thalidomide was used by pregnant women for the treatment of nausea. Later, it was discovered that thalidomide use resulted in phocomelia (a congenital disorder where the limbs of an exposed child are largely underdeveloped or absent). Before thalidomide use in pregnant women was banned, ~10,000 children were born with phocomelia. It is possible that a serious adverse effect of a medication, like that observed with thalidomide, would be noticed earlier today due to the increased use of OR.

It is important to remember that OR is best used to compliment traditional clinical trial research rather than replace it. The high internal validity of clinical trial research is often needed for the initial approval of a medication; yet the high applicability and external validity (i.e., reflecting real life) of OR is needed to build confidence in the utility of a medication in clinical practice.

OBJECTIVE 20.3. DEFINE AND EXPLAIN HOW TO MEASURE HEALTH-RELATED QUALITY OF LIFE AND PATIENT-REPORTED OUTCOMES

Patient preferences, along with evidence and clinical experience, make up the three core components of evidence-based medicine.[17] It is imperative to accurately measure and understand the patient perspective in both clinical practice and research. OR allows for the assessment of this perspective.

Defining Health-Related Quality of Life and Patient-Reported Outcomes

HRQoL is defined as *the impact of a health condition or treatment on the physical, psychological, and social aspects of life from the patient's perspective*.[18] An important concept in this definition is that HRQoL incorporates multiple aspects of life, not just the physical aspect (which is frequently the focus in current practice). HRQoL tools are commonly used in prospective trials to show the impact of a treatment on total well-being as it relates to health.

The US FDA defines PROs as *any report of the status of a patient's health condition that comes directly from the patient, without interpretation of the patient's response by a clinician or anyone else*.[18] Tools that can be classified as PROs may measure a patient's symptoms, functional status, or satisfaction with care.

Distinction Between Health-Related Quality of Life and Patient-Reported Outcomes

Although some use HRQoL and PRO interchangeably, others have argued that doing so is incorrect.[19,20] In this chapter, HRQoL and PROs will be presented as two distinct concepts. A key distinction between HRQoL and PROs is their scope.[20] Most HRQoL instruments or tools are completed by patients and are therefore PROs but not all PROs measure HRQoL. Tools used to measure HRQoL are generally broader than other tools used to measure PROs. As such, tools that measure HRQoL may not be sensitive to changes that other PRO tools could detect. On the contrary, some PRO tools may be too focused and thus not capture as many aspects of health as HRQoL tools. For example, a PRO may capture physical functioning, but not psychological or social functioning, whereas there are HRQoL tools available that measure all of these aspects. Currently, HRQoL and other PRO tools are most commonly used in prospective research studies. It is hoped, however, that with the increased recognition of these measures and the advent of technologies to increase access to them (e.g., electronic administration of the tools), PROs/HRQoL will play a bigger role in both routine care and research.[21]

Measuring Health-Related Quality of Life

HRQoL is measured through questionnaires, also known as instruments or tools, which are administered to patients.[22] The instruments consist of questions (also referred to as items) aimed at measuring patient experiences as a result of disease course. A group of these questions that fall under a common theme is referred to as a domain. Domains represent an aspect of health, with the most common domains being physical, psychological, and social functioning.

HRQoL instruments may be generic (i.e., those that can be used across various populations) or disease specific (Fig. 20.4).[22] Those that are classified as generic can be further subdivided into health profile measures and utility measures. *Health profile measures* attempt to capture all aspects of HRQoL. These generally ask patients a series of questions spanning multiple health domains, which are assigned a point value based on how the patient answers. A total score is then calculated either overall or within each domain. *Utility measures* incorporate patient preferences. With these measures, HRQoL is represented as a number between 0 and 1, with 0 representing death and 1 representing perfect health. The key advantage of using a generic instrument is that it allows for comparisons across disease states. For instance, a researcher could evaluate if a medication for asthma improved HRQoL to the same degree as a medication for heart failure. However, while disease-specific instruments make it more difficult to make comparisons across conditions, they are often able to detect smaller changes in the disease course than generic instruments are.

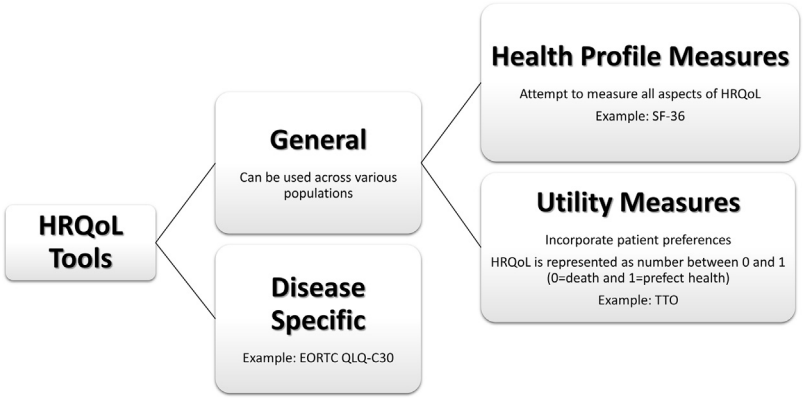


FIGURE 20.4 Categorization of health-related quality of life questionnaires. EORTC QLQ-C30 = European Organization for Research of Cancer Quality of Life Questionnaire Core-30; HRQoL = Health-related quality of life; SF-36 = 36-Item Short Form; TTO = time trade-off.

Generic Instruments: Common Examples

As mentioned above, generic health profile instruments attempt to capture all aspects of HRQoL.[22] A frequently used generic health profile instrument is the 36-Item Short Form (SF-36).[23] This questionnaire consists of the following eight domains: bodily pain, emotional role functioning, general health perceptions, mental health, physical functioning, physical role functioning, social role functioning, and vitality.[24] In addition to covering this comprehensive set of domains, the SF-36 has a minimal time burden associated with its use, as it takes ~7−10 min for participants to fill out the questionnaire.[23] Another common generic instrument is the abbreviated version of the World Health Organization Quality of Life Instrument (WHOQoL-BREF), which consists of 26 items and 4 main domains (i.e., physical health, psychological, social relationships, and environment).[25] In contrast, the WHOQoL-100 instrument has 100 items. The strength of the WHOQoL-BREF is that it is short and was specifically designed to be used across various cultures.

The EQ-5D-5L is a generic instrument that was developed by researchers from several countries in Europe.[26] It contains five dimensions (mobility, self-care, usual activity, pain/discomfort, and anxiety/depression) and assesses the following five levels of functioning: (1) no problems, (2) slight problems, (3) moderate problems, (4) severe problems, and (5) extreme problems. The EQ-5D-5L is available in over 130 languages and can be administered by paper or a variety of electronic devices.

Three commonly used health utility measures are the visual analog scale (VAS), time trade-off (TTO), and standard gamble.[27-32] The VAS is a simple measure, consisting of a line with 0 (representing death) on one end and 1.0 (representing perfect health) on the other.[28] Participants place a mark on the line to represent how they feel about their health state. The VAS has been criticized because it is not a real utility measure based on economic theory.

With the TTO, participants are given two options.[27] The first option is the health state (e.g., a disease state such as heart failure) for a set amount of time followed by death. The second option is living in perfect health for a set amount of time followed by death. Because living in perfect health is desirable, this second option is made shorter than the first. If the respondent prefers to live a shorter time but in perfect health, the questionnaire continues to make the time in perfect health shorter or longer until the participant would consider the two alternatives equal. For example, the instrument may read: would you rather live 40 years with heart failure or 38 years in perfect health? If the participant answers that he or she would rather live 38 years in perfect health, the next question would read: would you rather live 40 years with heart failure or 35 years in perfect health? Depending on the answer, this time in perfect health will then be increased or decreased. If the respondent prefers the 40 years with the disease, then the 35 years will again be increased to 36 years. As such, the time in perfect health would be decreased or increased until the participant finds the two options of equal value. At this point, the amount listed for the perfect health state is divided by the 40 years that was listed for the disease state, which gives a utility value (for instance, if the participant stated that they could not decide between spending 40 years with heart failure or 30 in perfect health, the resulting utility would be 30/40 or 0.75). The more time that the participant is willing to give up to live in perfect health rather than the reduced health state, the less favorably they view the health state. This will be reflected by lower utility values. Finally, the standard gamble produces a utility value by accounting for the probability of death an individual will theoretically accept to no longer experience the pain and suffering of their current condition.[29] Both the TTO and the standard gamble have been criticized due to their complexity.[30]

Disease-Specific Instruments

There are times when it is desirable to measure HRQoL in a way that is more focused on the impact of a particular disease.[33,34] Countless questionnaires have been developed for this purpose, including instruments such as the WHOQoL-HIV, Asthma Quality-of-Life Questionnaire (AQLQ), and the European Organization for Research and Treatment of Cancer Quality of Life Questionnaire Core-30 (EORTC QLQ-C30).[33,34] Disease-specific instruments allow for the inclusion of questions that are commonly used by clinicians to measure a particular disease. For instance, the AQLQ asks the patients questions about wheezing and shortness of breath that may not be applicable to other disease states.[33]

Reliability, Validity, and Responsiveness

When evaluating how well a HRQoL instrument performs, there are three concepts to consider: reliability, validity, and responsiveness.[23] *Reliability* refers to how consistent an instrument is. A tool with high reliability would yield consistent results when applied to similar patients under the same circumstances. *Validity* assesses if a HRQoL instrument adequately captures what it should measure. Finally, *responsiveness* refers to the ability of an instrument to detect important clinical changes in a patient's condition over time. If an instrument has high responsiveness, it is able to detect small changes in the disease course.

Measuring Domain-Specific Patient-Reported Outcomes

As previously stated, many PROs tend to focus on one domain, whereas HRQoL spans multiple domains. PROs may assess symptoms of disease, how well a patient can function (e.g., perform daily tasks such as walking or daily activities including caring for oneself), adherence to treatment, or satisfaction with care. Some PROs are mentioned in this section; however, a more extensive list of PRO measures can be found through the Patient-Reported Outcomes Measurement Information System (PROMIS).[35]

Symptom Measures

PROs that measure specific symptoms such as pain are common. These include scales like the one displayed in Fig. 20.5 or forms such as the brief pain inventory (BPI).[36] The BPI was designed to measure pain reported by cancer patients.[36] It asks patients about pain intensity and how this pain interferes with their life. The BPI is available in both a short (which consists of nine questions) and longer version.[37] Although pain is included in some of the above-mentioned HRQoL tools, it is only one of many domains. Because the BPI focuses on this one aspect, it may provide a more sensitive and detailed description of a patient's pain.

Adverse Event Reporting

When patients are asked questions to assess the presence of adverse events from a medication, this is a PRO.[38] In clinical trials assessing new medications, adverse event monitoring is required. Patients in these trials may be asked to answer questionnaires for the purpose of detecting adverse events that would otherwise be unknown to investigators. Adverse event reporting by the patients is common in cancer chemotherapy. Cancer clinical trials use the Common Terminology Criteria for Adverse Events (CTCAE) to describe the severity of adverse events.[36] CTCAE was initially designed for clinicians and thus contains medical jargon.[39] A revised version of CTCAE for patients uses simple language without losing its original content. For instance, instead of asking about "inadequate oral caloric or fluid intake" as was done in the original CTCAE, the patient-adapted version asks if they are "eating or drinking adequately.[40]"

Patient Satisfaction

PROs can also measure patient satisfaction with care. AHRQ's Consumer Assessment of Healthcare Providers and Systems (CAHPS) program is a predominant means of capturing patient satisfaction in the US.[41] The program administers standardized surveys that can be used to compare patients' experience with care across institutions. This is sometimes referred to as PREMs (Patient-Reported Experience Measures). It was originally designed to measure patients' satisfaction with their health plan but has been expanded to a spectrum of healthcare settings, including hospitals, nursing homes, and physician offices. Results from the CAHPS hospital survey (which asks the patient 32 questions, covering topics such as communication with healthcare providers, discharge instructions, and hospital cleanliness) are publically reported.

Patient or Clinician, Who Should Fill a Questionnaire?

Studies have shown that there is disagreement across questionnaires filled out by healthcare professionals about patient preferences compared with those filled out by the patient themselves.[42] This emphasizes that it is best to collect data about the patient experience directly from the patient. For instance, one study administered HRQoL questionnaires to both patients with prostate cancer and their physicians to fill out on behalf of the patient. The physicians underestimated patient impairment in all domains. Approximately 75% of patients reported fatigue, whereas physicians reported that fatigue was present in only 10% of patients ($P < .001$). Similarly, pain was reported by 43% of patients, but physicians reported that pain was present in <5% of patients ($P < .001$).[42]

Encouraging reporting of patient preferences is a method to improve patient-centeredness in clinical practice and research. This may increase the level of ownership patients take in their medical care. Engagement of patients in the clinical decision-making process is the cornerstone to establishing patient-centered clinical practice.[43]

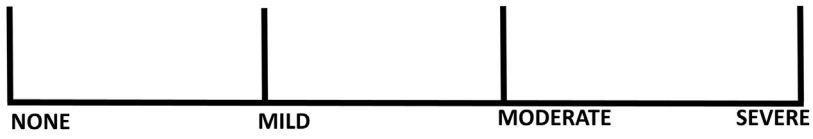


FIGURE 20.5 An example pain scale. A patient selects the vertical line that describes their pain intensity.

OBJECTIVE 20.4. GENERALIZE HOW OUTCOMES RESEARCH IS GAINING IMPORTANCE IN CLINICAL PRACTICE

The cost of healthcare is increasing, and there is a push toward reimbursement of services based on value not volume. Value is subjective. Value can be measured from a patient's perspective (e.g., value = what did the patient get—how much money the patient is willing to pay to get it) versus a healthcare system's perspective, which takes into account the outcomes, the harm, and the resources consumed.[44] A primary concern for healthcare systems is how the insurance companies and payers define value because this determines how the healthcare organization is reimbursed. Each institution is also compared against other like institutions, so if one organization achieves a better perceived value, then other institutions will be forced to follow. Centers are allocating more and more resources into looking at center-specific outcomes and improving value, thus creating an opportunity for OR.

Scenario 1: Outcomes Research in Medical Reimbursement Decisions

Historically, a hospital was paid a certain amount of money as medical reimbursement for performing a surgery to remove the appendix. The operation and length of stay would be covered within reason. Now, depending on how well a surgeon performs this operation (based on their complication rate and expected length of stay for patients), reimbursement between payers may change, even if they perform the same operation at the same center. That means if a patient has to stay an extra 3 days because of a wound infection, the hospital may not get reimbursed for the full expense of that 3 extra days. This is an alternative payment model (APM) or value-based purchasing (VBP) model. These are models in which third-party payers either provide incentives or penalties for measures based on quality and/or costs and may motivate improvement.[45]

Changes in the US demonstrate the refocus of healthcare reimbursement on quality and value versus traditional fee for service. For example, a Merit-based Incentive Payment (MIPS) system is a relatively new term. These came about with the Medicare Access and CHIP (Children's Health Insurance Program) Reauthorization Act of 2015 and are an artifact of the Patient Protection and Affordable Care Act.[46] This is an example of a push for Medicare, the largest provider of public healthcare in the US, to reimburse physicians based on cost and "value" versus the traditional fee for service.[46] The MIPS program provides a penalty or a bonus based on the performance measures and encourages providers to use technology and make improvements while containing costs and optimizing quality.[46] Several other countries are exploring different APM or VBP models trying to figure out which among these will work better: incentives, penalties, or bonuses. This concept is not new in the pharmaceutical industry. Health plans negotiate with pharmaceutical companies to define tiers for medications.[47] This may incentivize medication manufacturers to provide their medication at a lower cost when compared with another medication within the same class or approved indication. OR is essential for measuring value in healthcare and is used widely to help in reimbursement decisions.

Scenario 2: Interpreting OR Data in a Particular Context

The OR data behind metrics should be understood. Suppose a hospital wants to analyze the impact of medications used for empiric coverage of urosepsis because the mortality index for the hospital is increasing. How should this be approached? It is first important to understand the mortality index and how this is measured. Specifically, in looking at the medications used for empiric coverage of urosepsis, the index values are usually based on an equation containing observed mortality over expected mortality. The expected outcome is determined by a model. For example, a previously healthy 30-year-old person presenting to the emergency room with urosepsis has a lower risk of mortality versus an 87-year-old coming from a nursing home with urosepsis. It is important to identify the patients falling into the model and what variables are predicting a higher rate of mortality. In this particular context, the hospital may discover that all of the patients who are dying from urosepsis are not patients who received the wrong empiric medication, but rather they were patients who were underdosed or had a delay in receiving medication. This then makes the answer to the question posed entirely different. Important take-home points when assessing score reports in a particular context are as follows: (1) take a look at the data behind the scores, (2) what methods and definitions are used and what do they mean, (3) what is most relevant (e.g., is a metric that tries to capture overuse of medical imaging or a metric that captures wait time in the emergency room most relevant to patients?), (4) when was the data obtained (data from 10 years ago may not be applicable today), and (5) what is the methodology for reporting (e.g., ranking based on a survey of other physicians or actual data). Thus, score reports should not be interpreted alone. Rather, one should understand how these score reports are calculated to make appropriate clinical decisions.

Scenario 3: Outcomes Research Data in a Healthcare System Performance

OR can be used to analyze the performance of healthcare systems. A classic example is the solid organ transplant services in the US. In January of 2017, the Centers for Medicare and Medicaid services (CMS) adopted new flagging criteria for outcomes, specifically 1-year graft loss and patient mortality.[48] What does that mean? Every 6 months the Scientific Registry of Transplant Recipients releases a transplant program-specific report. This report details all hospital transplants during the reporting period and provides an observed number of graft losses and patient deaths compared with expected number. If a center experiences a higher number of deaths or graft losses than expected and it exceeds some other particular thresholds set by CMS, then the center is at risk of losing funding. This is especially true if the center cannot explain why the observed number of graft losses and patient deaths is higher than expected or explain steps that the center has taken to make improvements and subsequently demonstrate improvements on upcoming reports. If a center loses CMS funding for the transplant program, most other insurance payers will follow suit and also pull their funding. Many third-party health plans only cover centers covered by Medicare. Here is the challenging part! In the case of transplant outcomes, a center is assessed based on 2.5 years' worth of data. So when the next report is released in January of 2018, the data that will be evaluated for 1-year risk of graft loss or mortality are based on outcomes from patients who received transplants from July 1, 2014 through December 31, 2016. This is likely implemented to ensure that most patients have arrived at their 1-year milestone, but the data lag does not demonstrate what may be going on now. Some examples include a center with a new surgical team that was not involved in transplanting the patients included in the data or the center was shut down for 9 months following a hurricane so the denominator is smaller. It is also important to understand what goes into the model. Some mortality models, for example, exclude patients who have had a previous transplant for the organ of interest. Furthermore, a pharmacist member of the team where outcomes are not looking great may be able to suggest an alteration in the immunosuppression regimen used by the center.

OR, along with other measures, is thus useful in analyzing healthcare system performance. However, one must always understand how and in what context data were obtained. For instance, it is known that the infant mortality rate (an OR indicator) is used to measure system performance in the facility, regional, national, or international levels. However, although this measure is universal, it would be unwise to make a decision without accounting for where and how this measure is calculated.

CONCLUSION

In summary, OR places emphasis on patient perspectives and represents multiple stakeholders in healthcare. The most conclusive way to evaluate how a healthcare system functions is by measuring the end result of the interventions. This increases the need for OR to generate real-world safety and effectiveness data. Use of HRQoL and other PRO tools has value in both OR and clinical trials. Patient reporting also improves patient-centeredness in clinical practice. OR is an important tool for improving healthcare across various settings.